Transparent Saliency Routing for Temporal Image Drift

A compact mock research paper • fixed-horizon prototype run • 29 September 2026

Abstract

This prototype evaluates a ViT-only drift correction pipeline whose update decision can be inspected directly. Historical models train on 2004–2008, model selection uses 2009, drift adaptation uses 2010, router calibration uses 2011, and 2012–2014 remain untouched until final evaluation. Under combined drift in 2014, the transparent router improves macro F1 by 1.16 percentage points over the historical XAI model and retains 82.3% of the gain available from always choosing the updated model.

1. Research question

Can visible saliency change and prediction evidence determine when a drift-adapted ViT should replace a historical ViT prediction across a realistic future horizon?

2. Architecture and transparent routing

The experiment compares a baseline ViT, an attention-regularised XAI ViT, and a paired historical/newer XAI system. Its router is a standardized logistic rule over four logged quantities: saliency Jensen–Shannon divergence, confidence gain, entropy reduction, and class-evidence gain. A minimum saliency-change floor must also be passed. The saved coefficients and threshold make every update decision reproducible.

2.1 Model mechanics

DINOv2-small emits a class token and patch-token grid. The baseline pools patches uniformly. The XAI model learns one gate per patch, normalises the gates into a saliency distribution, and concatenates the saliency-weighted patch sum with the class token before classification. The explanation therefore participates directly in prediction.

Component	Mechanism	Role in drift correction
DINOv2-small	Class token + patch tokens	Shared ViT representation
Baseline head	Uniform patch pooling	Historical reference
XAI saliency head	Learned scalar gate per patch	Weights patches inside forward pass
Counterfactual path	Complement of salient patches	Tests class-evidence faithfulness
Twin comparison	Frozen historical + adapted newer ViT	Measures prediction and map change
Transparent router	Four-feature logistic rule	Chooses old or new prediction

2.2 Saliency faithfulness and update loss

A complementary path pools low-saliency patches. Training penalises the model unless selected patches provide greater true-class evidence than their complement. During adaptation, the pairwise term keeps historical and newer maps close when the historical prediction is correct, while imposing a divergence margin when it is wrong. Classification, pairwise saliency, and counterfactual evidence losses are optimized jointly.

2.3 Routing equation

For saliency JS divergence, confidence gain, entropy reduction, and evidence gain, the router computes z = (f – mean) / scale and p(update) = sigmoid(w·z + b). It selects the newer class only when the models disagree, saliency JS clears the saved floor, and p(update) clears the saved threshold. Every term is logged for sample-level audit.

3. Chronological protocol

The split models deployment: historical learning stops before adaptation, and the future horizon is used only for final measurement. This prevents year-by-year retraining from leaking test information into the model.

Stage	Years	Purpose
Historical training	2004–08	Fit original ViTs
Selection	2009	Choose checkpoints
Adaptation	2010	Update XAI ViT with mild drift
Calibration	2011	Fit explicit route rule
Frozen test	2012–14	Measure future drift

3.1 Optional realistic drift

The drift toggle applies mild saturation change, Gaussian sensor noise, and partial occlusion. The newer ViT sees a deterministic 50/50 mixture of clean and individually corrupted 2010 samples. The router is calibrated on moderate individual corruptions in 2011. Combined corruption is withheld until 2012–2014, making it a composition shift rather than a memorised augmentation recipe.

3.2 Saved routing policy

Transparent feature	Coefficient
Saliency Js	+0.240
Confidence Gain	+0.440
Entropy Reduction	-0.321
Evidence Gain	-0.055
Route probability threshold	0.35
Saliency-change floor	0.0022

Coefficients act on standardized features. Their signs describe the 2011 calibration relationship; they are not causal effects. The explicit saliency floor ensures that saliency remains part of the decision rather than a decorative post-hoc plot.

4. Enhanced correction experiment

Status: implemented in commit 42ca356; results pending because the remote server was unavailable at launch. The 3-point value below is a predeclared target, not an observed result.

Controlled item	Standard profile	Enhanced profile
2010 stream	50% clean; severity 1	25% clean; individual severity 1–3
Trainable backbone	Final ViT block	Final three ViT blocks
Learning rates	All trainable: 1e-5	Backbone 1e-5; heads 5e-5
Label smoothing	0.00	0.05
Saliency loss weight	0.10	0.25
Divergence margin	0.05	0.08
Saliency floor	Fixed median	2011-calibrated quantile

4.1 Fixed controls

Historical checkpoint, CLEAR-100 chronology, random seed, epoch budget, 2011 calibration labels, frozen 2012–2014 test streams, corruption code, batch size, and metrics must match the standard run. Combined corruption remains absent from adaptation and calibration.

4.2 Predeclared hypotheses

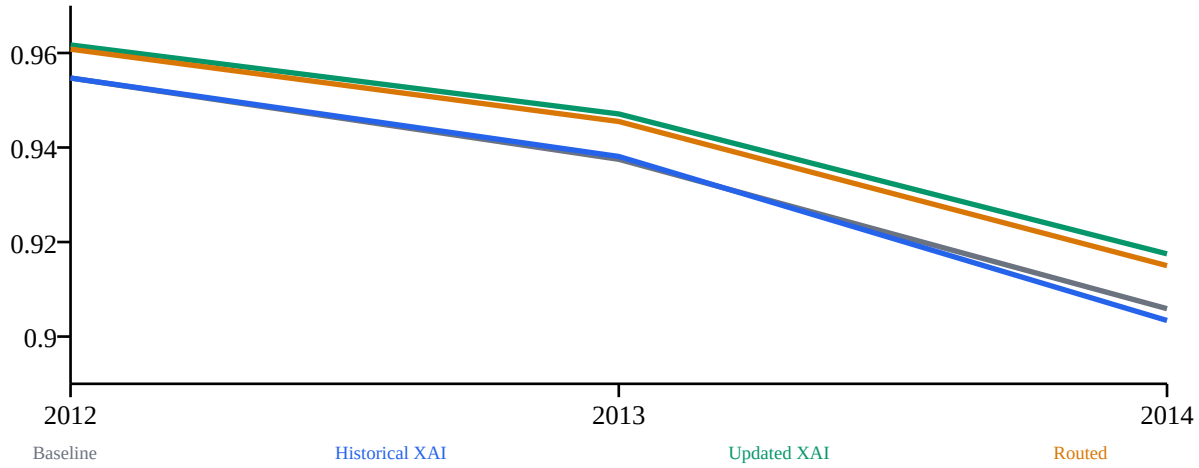
H1: at least +3.00 macro-F1 points over baseline ViT on 2014 combined drift. H2: routing retains at least 75% of the always-new gain. H3: 2014 clean macro F1 falls by no more than 0.50 points versus historical XAI. Missing H1 is a negative result; test labels must not be used for another tuning cycle.

4.3 Evaluation matrix

Report baseline, historical XAI, updated XAI, and routed predictions for every year and condition. Primary metric: macro F1. Secondary metrics: accuracy, macro IoU, update-trigger precision/recall/F1, helpful and harmful routes, training seconds per epoch, inference milliseconds per image, and saliency-map overlap. Repeat across seeds before making a final claim.

5. Completed fixed-horizon results

Macro F1 under combined saturation, noise, and occlusion. The widening decline across years makes the temporal effect visible.



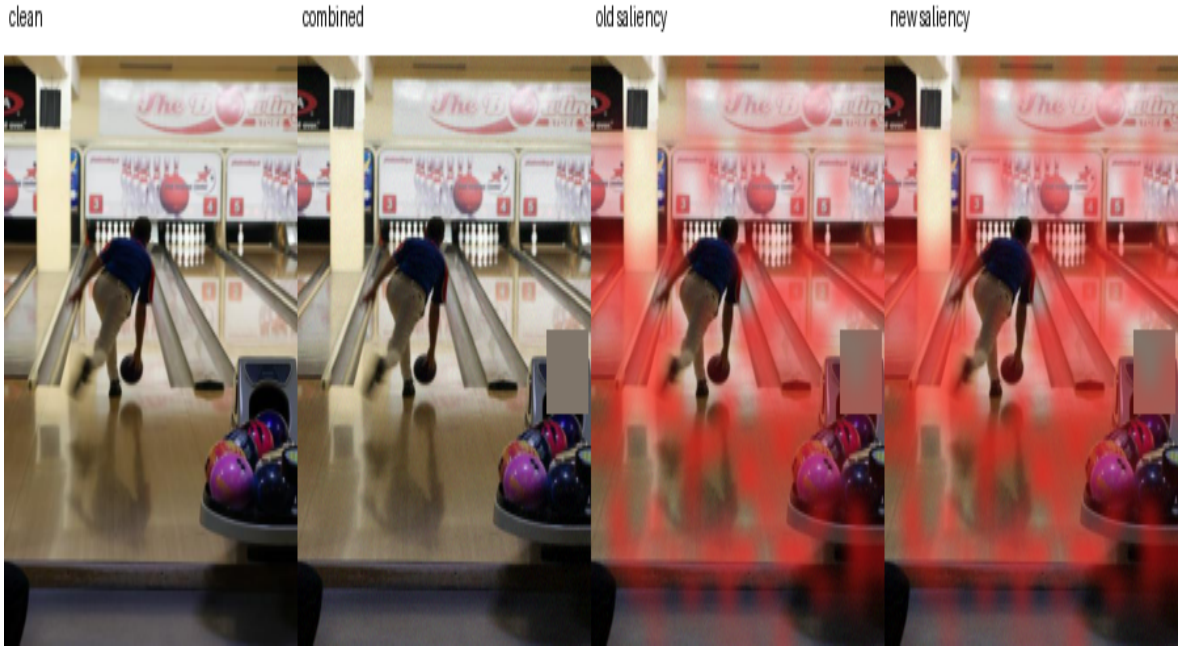
Year	Baseline	Hist. XAI	Updated XAI	Routed	Helpful / harmful
2012	95.47	95.47	96.17	96.08	45 / 14
2013	93.75	93.81	94.71	94.55	72 / 32
2014	90.59	90.34	91.75	91.50	120 / 48

In 2014, the updated XAI model gains 1.41 points over historical XAI. Routing retains 1.16 points: (91.50–90.34)/(91.75–90.34) = 82.3% of the available gain. Clean-data macro F1 is 95.13% baseline, 95.24% historical XAI, 95.85% updated XAI, and 95.46% routed.

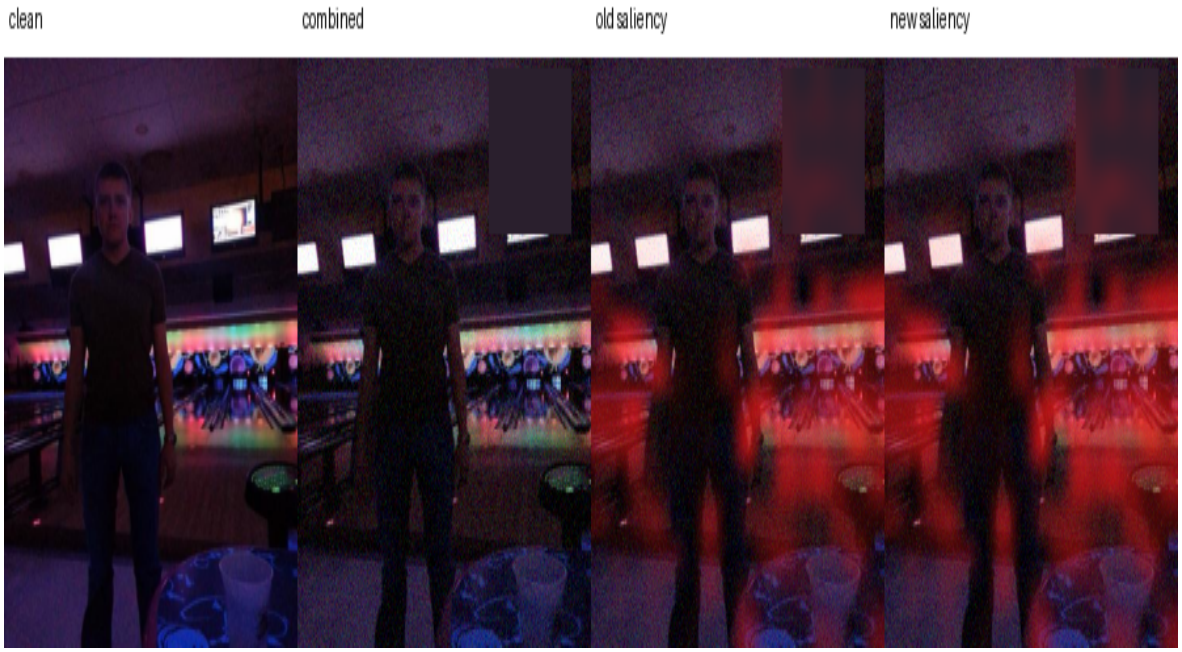
6. Saliency audit examples

Each panel shows the clean reference, drifted input, historical saliency, and updated saliency. Together with the saved decision log, this lets a reviewer connect the route to visible attention changes and prediction evidence.

Combined drift • 2012



Combined drift • 2014



7. Interpretation

Compound drift produces a clearer gap than the earlier one-year test. The completed standard run shows that the adapted ViT improves drifted and clean predictions and that routing captures most of the improvement. Confidence gain has the strongest positive learned coefficient; saliency divergence contributes positively and is reinforced by the explicit saliency floor. Coefficient signs are calibration correlations rather than causal explanations.

8. Limitations and validity threats

The completed evidence is one exploratory seed on one temporal stream with synthetic corruptions and one calibrated threshold. It has no confidence intervals or external validation dataset. The new 3-point target is an experimental objective, not a reported achievement. Increasing adaptation capacity could improve accuracy because of added optimization capacity rather than saliency alone, so ablations must isolate trainable blocks, corruption curriculum, saliency loss, and router calibration.

9. Next experimental sequence

Run standard and enhanced profiles with identical controls; freeze outputs; compute paired per-sample bootstrap intervals; then repeat across at least three seeds. Compare always-old, always-new, confidence-only, saliency-only, and full routing. Finally, repeat the untouched protocol on a second temporally indexed image dataset before claiming generality.

10. Reproducibility

The repository includes a compact JSON summary, learned router policy, 45 saliency panels, report generator, and all training source. Large checkpoints remain server-side. Completed run identifiers: baseline 20260928T152657Z-horizon, historical XAI 20260928T161159Z-horizon-xai, and router 20260928T161159Z-horizon-router. Enhanced-profile results remain pending.

References

[1] Dosovitskiy et al., Vision Transformer, 2020. [2] Oquab et al., DINOv2, 2023. [3] Hendrycks & Dietterich, ImageNet-C, 2019. [4] Hendrycks et al., AugMix, 2019. [5] Lin et al., CLEAR, 2021. Full links appear in the Markdown version.